

Life Expectancy Changes Due to Kidney Donation Across Race, Sex, and Age

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Abstract

Background. Living kidney donors face a small but well-documented elevation in long-term risk of end-stage renal disease (ESRD), yet this risk has not been translated into a decision-relevant life-expectancy metric stratified by the demographic factors known to modify it.

Methods. I constructed a Markov microsimulation model comparing lifetime outcomes for matched donor and non-donor cohorts, entering at the age of donation and simulated in annual cycles, parameterized from USRDS, SRTR, and published cohort studies of post-donation ESRD incidence and mortality.

Results. In the base case, donation was associated with a mean life-expectancy cost of 37 days. This cost was nearly three times larger for Black donors than White donors and more than three times larger at age 25 than at age 55. Sensitivity analyses identified the donor all-cause mortality hazard ratio, for which the literature reports conflicting point estimates, as the dominant source of uncertainty, dwarfing every other modelled parameter. The life-expectancy value of prior-living-donor priority waitlist access and of a National-Kidney-Registry-style donation voucher was negligible unconditionally (under two days) but worth several hundred days conditional on the recipient's actually developing ESRD.

Conclusions. These findings support demographic-specific life-expectancy disclosure during living-donor informed consent and indicate that priority-access mechanisms function as catastrophic insurance rather than as interventions expected to shift average outcomes.

Keywords

living kidney donation, life expectancy, Markov model, end-stage renal disease, health disparities

Highlights

- Kidney donation costs the average donor about five weeks of life expectancy.
- This cost is nearly three times larger for Black donors than White donors, and more than three times larger at age 25 than at age 55.
- The biggest source of uncertainty is donor hazard ratio.
- Priority transplant access for donors and their designated family members is worth almost nothing on average, but hundreds of days if kidney failure actually occurs.

Introduction

Living donor kidney transplantation (LDKT) remains the treatment of choice for end-stage renal disease (ESRD), offering better graft function and longer allograft survival than deceased-donor transplantation.¹ Yet the gap between organ need and organ supply persists: more than 4,000 candidates died on the US kidney waiting list in 2024, and a comparable number were removed for transplant-disqualifying deterioration, while the annual number of LDKTs has remained essentially flat for over a decade.² Expanding living donation therefore rests on two complementary levers: (1) recruiting more donors and (2) better understanding, quantifying, and managing the risk that donation poses to the donor.

The safety literature on living donation is large, but has historically emphasized relative rather than absolute, or life-expectancy-scaled, risk. Perioperative mortality is low (3.1 per 10,000 donors) but not uniform, rising sharply with male sex, Black race, and pre-existing hypertension.³ Longer-term follow-up using healthy, contraindication-screened non-donor comparators, rather than the general population, established that living donors face a several-fold higher risk of subsequently developing ESRD themselves, even though the absolute magnitude of that risk remains small.^{4,5} Whether this translates into excess all-cause mortality is unresolved: a meta-analysis of 52 studies and over 118,000 donors found no elevation in all-cause mortality relative to matched non-donors,⁶ whereas a Norwegian registry study with unusually long follow-up and a cohort drawn predominantly from first-degree relatives of recipients reported a modestly elevated hazard ratio of 1.30.⁷ This tension between the largest available meta-analysis and

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the longest single-cohort follow-up is a central source of uncertainty for any model of donor life expectancy.

Post-donation ESRD risk is not distributed evenly across the donor population. Black donors experience roughly three times the 15-year cumulative incidence of ESRD observed in White donors, a disparity that persists after matching to equally healthy non-donor comparators of the same race.⁴ Part of this excess risk has a defined biological correlate: two-copy *APOL1* renal-risk-variant genotypes, disproportionately carried by individuals of recent African ancestry, are among the strongest known risk factors for progression to CKD stage 3 or higher and have been directly implicated in the long-term kidney health of Black living donors.⁸ Donation itself is also not evenly distributed: women have consistently made up a majority of living donors, a pattern attributable in part to financial barriers to donation and disparities in access to unpaid leave that fall more heavily on lower-income men.⁹ Together, these findings motivate stratifying any life-expectancy analysis of donation by age, race, and sex rather than reporting a single population-average effect.

Policy has responded to donor risk primarily by protecting donors' future access to a transplant should they themselves develop ESRD. The Kidney Allocation System implemented in 2014 (KAS250) grants prior living donors priority access to the deceased-donor waiting list, and early evaluation confirmed that this priority meaningfully shortened prior donors' time to listing.¹⁰ Independently, a national registry analysis of transplant candidates who had previously donated an organ found that such priority access translated into substantially shorter waiting times and higher-quality kidneys compared with matched non-donor candidates, confirming that the policy achieves its intended effect at the level of the individual recipient.¹¹ The National Kidney Registry has extended this priority concept beyond the donor's own future risk through its voucher program, under which a donor may designate family members, or in some cases any future intended recipient, to receive expedited, priority access to a living-donor kidney should they ever need one. Two large Monte Carlo simulation studies, spanning 50-year and, subsequently, 100-year horizons, have shown that the voucher program is programmatically sustainable: under a wide range of donor-growth scenarios, the supply of future voucher-eligible donors is projected to exceed voucher redemptions by a comfortable margin.^{2,12}

These two literatures, quantifying the donor's own excess ESRD and mortality risk, and demonstrating the programmatic feasibility of priority-access mechanisms such as KAS250 and the NKR voucher, have so far remained largely separate. The voucher simulations answer a supply-side question (will there be enough donors to honor future redemptions?) but do not quantify the demand-side question of how much life expectancy a voucher, or PLD priority access more generally, is actually worth to the person who holds it. Similarly, the ESRD-risk literature reports relative risks and cumulative incidences but stops short of translating these into an integrated, individual-level life-expectancy trade-off that simultaneously accounts for the donor's elevated ESRD risk, the mortality and quality-of-life consequences of dialysis and transplant waitlisting, and demographic heterogeneity. This paper addresses that gap

using a Markov microsimulation model that estimates the change in life expectancy attributable to kidney donation itself, stratified by age at donation, race, and sex, and separately quantifies the life-expectancy value of priority waitlist access, both unconditionally at the population level and conditional on the recipient having actually developed ESRD.

Methods

Model overview

I constructed a Markov state-transition model to compare lifetime outcomes for two hypothetical cohorts of otherwise identical individuals: one cohort that donates a kidney and one that does not. Both cohorts enter the model at the age of donation in full health, and the simulation advances in annual cycles until age 110. The analysis was conducted from the perspective of the individual donor rather than the health-care system; outcomes are reported as undiscounted remaining life-years, consistent with the convention in burden-of-disease analyses, which discontinued time discounting so that a life-year counts equally regardless of when it occurs.¹³ The base-case age at donation is 40 years; subgroup analyses span ages 25–55. I used a Monte Carlo simulation with 5×10^6 individuals per arm and cross-validated against a deterministic analytic cohort implementation. I used annual (one-year) cycles throughout. The Markov model structure is shown in Figure 1.

Health states

The model contains seven mutually exclusive health states (Figure 1): *Healthy*, *ESRD onset*, *Dialysis*, *Standard waitlist*, *Priority waitlist*, *Post-transplant*, and *Dead* (absorbing). At ESRD onset, the cohort branches: a fraction is listed preemptively (bypassing dialysis entirely) and the remainder initiates haemodialysis. The donor arm routes ESRD survivors to the *Priority waitlist*, reflecting the allocation priority afforded to prior living donors (PLD) under KAS250; the non-donor arm routes to the *Standard waitlist*. Graft failure returns patients to the *Dialysis* state, bypassing the ESRD-onset branch.

ESRD risk after donation

Fifteen-year cumulative ESRD incidence was taken from Muzaale et al.⁴: 30.8 per 10,000 (95% CI 24.3–38.5) for donors and 3.9 per 10,000 (95% CI 0.8–8.9) for age-, sex-, and health-matched controls (8-fold relative risk). Race-specific donor rates were 74.7/10,000 (Black) and 22.7/10,000 (White).⁴ The White non-donor baseline was taken from Grams et al.¹⁴ (sex-averaged 0.05%/15 yr), because zero ESRD events were recorded in White matched controls in Muzaale et al.

I modelled ESRD hazard with a Weibull distribution (shape $k = 1.5$), calibrated via a competing-risk bisection algorithm to reproduce the 15-year cumulative incidence while accounting for background-mortality attrition in the at-risk pool. The accelerating hazard ($k > 1$) is supported by the Massie et al. 20-year cumulative incidence of 34 per 10,000 (vs. $\approx 21/10,000$ predicted under a constant-hazard

model).⁵ I drew within-donor subgroup hazard ratios from Massie et al.: 2.96 (95% CI 2.25–3.89) for Black race, 1.88 (95% CI 1.50–2.35) for male sex, and 1.40 per decade (non-Black donors).⁵

Dialysis survival and waitlist listing

First-year haemodialysis mortality was 22% and subsequent-year mortality 17% ($\approx 42\%$ five-year survival), consistent with USRDS estimates.¹⁵

At ESRD onset, a fraction of patients is listed preemptively before dialysis initiation, avoiding dialysis mortality entirely. I drew preemptive listing probabilities from USRDS 2025 ADR Figure 7.13: 5.8% overall for the non-donor arm and 9.4% for the donor arm, using the age 18–44 population rate as a proxy for this younger, healthier, donor-like cohort (USRDS does not report preemptive-listing rates stratified by donor status).¹⁶

Among patients on dialysis, the conditional annual probability of joining the waitlist was set to 15%/year (base case), derived from USRDS 2025 ADR Figure 7.17.¹⁶ Approximately 25% of 18–44 year-old ESRD patients are listed or transplanted within one year, of whom $\approx 9.4\%$ were preemptively listed, yielding an estimated post-dialysis listing rate of $\approx 15.6\%$ /year for the donor-like cohort (conservative base case: 15%/year; one-way sensitivity range 5–30%/year).

Transplant waiting time and waitlist outcomes

Transplant probability while listed was modelled as a time-homogeneous exponential process with rate $r = 1/\mu$, where μ is the mean waiting time, the natural parameterisation of the exponential distribution.¹⁷ For the standard waitlist, I used the post-KAS250 national mean of 1,765 days (Punjala et al. 2024, Table 3).¹⁸ For the PLD priority waitlist, I used the post-KAS median of 102.6 days reported by Wainright et al.,¹⁰ converted to a mean via $\mu = \text{median}/\ln 2 \approx 148$ days. Waitlist mortality was 5.0 deaths per 100 patient-years (SRTR 2023 ADR Figure KI 24). Competing removal (delisting for reasons other than transplant or death) was derived from the 3-year cumulative removal incidence of 19.06% reported in SRTR 2023 ADR Figure KI 22, back-calculated to a cause-specific annual rate of 11.4%/year via competing-risk bisection accounting for simultaneous attrition from transplantation and death.¹

Post-transplant survival

I derived post-transplant annual mortality from age-stratified 5-year Kaplan–Meier estimates for DDKT recipients in the SRTR 2023 ADR (Figure KI 70, 2016–2018 cohort): 0.9%/year (ages 18–34), 1.8%/year (35–49), 3.9%/year (50–64), and 6.9%/year (≥ 65).¹ The base case uses DDKT survival because prior living donors who develop ESRD predominantly receive transplants via the deceased-donor priority pathway rather than living-donor transplantation, consistent with USRDS data showing that living-donor transplants account for a small minority of transplants in this population.¹⁹ A one-way sensitivity analysis substitutes LDKT patient survival (SRTR 2023 ADR Figure KI 76: 0.4%, 0.8%, 1.7%, and 3.9%/year by age band) to bound the effect of this assumption. Annual graft failure post-year 1

was likewise age-stratified, derived as $1 - (S_5/S_1)^{1/4}$ from age-specific 1- and 5-year DDKT graft survival in the SRTR 2023 ADR (Figure KI 53, 2016–2018 cohort): 3.9%/year (ages 18–34), 3.4%/year (35–49), 4.9%/year (50–64), and 7.8%/year (≥ 65).¹ Graft failure returned patients to the Dialysis state.

Background mortality and donor all-cause survival

I took background mortality in the Healthy state from the 2021 US Life Tables,²⁰ with sex-specific tables used in sex-stratified analyses. The base-case donor all-cause mortality hazard ratio versus matched non-donors is 1.0, consistent with US-based studies^{3,4} and the O’Keeffe et al. 2018 meta-analysis (pooled RR 0.60, 95% CI 0.31–1.10; 4 studies, $n = 84,495$ donors).^{6,21} I also examined an HR of 1.30 from Mj  en et al.⁷, derived from a Norwegian cohort in which 80% of donors were first-degree relatives of the recipient, as the pessimistic sensitivity scenario.

Subgroup analyses

Age at donation I repeated the analyses at ages 25, 35, 40, 45, and 55 years at donation. Donor ESRD rates were scaled by the Massie 2017 age HR (1.40/decade, non-Black donors);⁵ I held non-donor baselines at the Muzaale overall matched-control rate.

Race I took race-specific donor ESRD rates (Black 74.7/10,000; White 22.7/10,000) and race-specific non-donor baselines (Black sex-averaged 0.195%/15 yr; White sex-averaged 0.05%/15 yr) from Muzaale et al.⁴ and Grams et al., respectively.¹⁴ Race-specific waitlist mortality and post-transplant survival were from SRTR 2023 Figures KI 25 and KI 71.¹

Sex I derived sex-specific donor ESRD rates were derived by applying the Massie 2017 male-sex HR (1.88)⁵ to the overall donor rate, weighted by the observed donor sex mix (63.5% female; SRTR 2023 ADR Figure KI 104).¹ For non-donor baselines, we used the Grams 2016 White sex-specific rates (female 0.04%/15 yr; male 0.06%/15 yr)¹⁴ rather than applying the within-donor sex HR to the control arm, because Massie 2017 hazard ratios describe variation within the donor cohort and are not appropriate for scaling a different reference population. I applied this sourcing rule, Muzaale matched-control rates for overall analyses and Grams direct rates wherever sex or race is stratified, consistently across all simulations.

Probabilistic sensitivity analysis

I conducted a PSA with 200 parameter draws. Sampled quantities and their distributions were:

- ESRD 15-year cumulative risks: beta distributions parameterised from the Muzaale 2014 published 95% CIs;
- Donor all-cause mortality HR: log-normal, $\mu = -0.016$, $\sigma = 0.143$, derived from the Grams 2018 meta-analysis CI;
- Weibull shape k : log-normal, $\sigma = 0.13$, representing structural uncertainty absent published CIs for this parameter; and

- Waitlist listing probability: beta, fractional SE = 40%, consistent with the 0.05–0.30 plausible range

Registry-derived parameters (from SRTR or USRDS) have negligible sampling error relative to structural uncertainty and were fixed at base-case values; scenario variation in these parameters is captured by the one-way sensitivity analysis. I report the mean ΔLE , 95% credible interval, and the probability that donation is life-expectancy–neutral or beneficial.

One-way sensitivity analysis

I varied each parameter individually:

- No priority access and changing the PLD wait time
- Donor all-cause mortality HR = 1.30 from the Mj  en pessimistic estimate
- ESRD relative risk from the Mj  en upper-bound on ESRD HR
- Post-transplant survival at LDKT quality
- Dialysis mortality $\pm 50\%$

Calibration

I validated the model’s transplant probability against the SRTR 2023 3-year waitlist outcome distribution (Figure KI 22, 2018–2020 listing cohort).¹ This comparison is non-circular because I derived the transplant probability from the mean waiting time, not from the 3-year outcome distribution. I back-calculated the waitlist removal rate via a competing-risk bisection algorithm to reproduce the SRTR-observed 19.1% removed at three years (11.4%/year); the na  ve formula $1 - (1 - \text{CIF})^{1/3}$, which ignores competing depletion by transplantation and death, was not used. The model over-predicted three-year waitlist mortality (10.5% vs. 6.8% observed, $\Delta = +54\%$), likely reflecting the application of a 2023 cross-sectional mortality rate (confirmed 5.0/100 patient-years from the raw SRTR data file) to a 2018–2020 listing cohort; this bias is symmetric across both arms and conservative (slightly overstates the cost of the waitlist period for both donors and non-donors equally). The model predicted 38.3% transplanted at three years versus 44.3% observed (DDKT + LDKT combined). No independent cohort exists tracing the complete pathway from living donation through ESRD onset to transplantation; this check represents the available face-validity evidence for the model.

Voucher benefit

A kidney-donation “voucher” is a mechanism by which a living donor designates a healthy, non-donor family member to receive PLD priority-waitlist access should that family member ever develop ESRD. I simulated two arms of healthy non-donors, both carrying non-donor baseline ESRD risk and a background all-cause mortality HR of 1.0; the sole difference between arms was waitlist priority upon ESRD onset (voucher arm: PLD mean wait 148 days, derived from the Wainright et al. post-KAS median of 102.6 days via $\mu = \text{median} / \ln 2$;¹⁰ control arm: standard mean wait 1,765 days¹⁸). Consistent with the sourcing rule used for the base-case preemptive-listing parameters, I assigned the voucher arm the informed/monitored preemptive-listing probability (9.4%) and the control arm the

general-population probability (5.8%).¹⁶ All other transition probabilities (dialysis mortality, per-cycle waitlist listing, waitlist mortality and removal, post-transplant survival, graft failure) were identical to the base-case model. I ran the analysis at the base-case age of 40 with an age sweep (25–60 years) and an ESRD-risk subgroup analysis using the Muzaale overall and Grams race-specific non-donor CIFs.^{4,14} One-way sensitivity analyses varied the voucher and standard median wait times, the per-cycle listing probability, the preemptive-listing probability, and post-transplant graft quality (DDKT vs. LDKT).

ESRD-conditional priority benefit

Because the voucher (and, by extension, the donor PLD-priority) benefit is diluted at the population level by the large fraction of the cohort who never develop ESRD, I ran a second model conditioned entirely on ESRD onset. All individuals entered the model already diagnosed with ESRD in order to isolate the life-expectancy benefit of priority waitlist access for the sub-population who actually reach the waitlist. Entry to dialysis versus preemptive listing, dialysis and waitlist mortality, waitlist removal, post-transplant survival, and graft failure used the same age-stratified, all-cause parameters as the base-case model; no separate Healthy state or background-mortality term was included, since dialysis- and waitlist-state mortality rates are all-cause rates already measured in ESRD populations. I compared two arms, differing only in waitlist priority (priority vs. standard median wait, as above). The base case used an ESRD-onset age of 60 with an age sweep (40–75 years at ESRD onset) and one-way sensitivity analyses on wait times, per-cycle listing probability, preemptive-listing probability, post-transplant graft quality, and dialysis mortality ($\pm 20\%$).

Results

Base-case life expectancy

In the base-case analysis (age 40 at donation, overall population parameters), kidney donation was associated with a life-expectancy cost of $\Delta\text{LE} = -40.5$ days (donor 38.04 yr vs. non-donor 38.15 yr remaining life expectancy; Monte Carlo, 5×10^6 individuals per arm). The analytic cohort confirmed this estimate ($\Delta\text{LE} = -40.4$ days). Across all base-case and subgroup scenarios, donation was associated with a net life-expectancy cost to the donor; no scenario identified donation as neutral or beneficial to the individual.

Racial disparity

Race was the strongest subgroup modifier of life-expectancy cost (Figure 2). Black donors experienced a cost of 81.2 days, three times the 27.2 days estimated for White donors at age 40. The disparity widened sharply with younger age at donation: a Black donor at age 25 loses 212 days of life expectancy compared with 37 days for a White donor of the same age. This disparity traces directly to post-donation ESRD incidence: Black donors have a 15-year cumulative incidence of 74.7 per 10,000 versus 22.7 per 10,000 in White donors,⁴ a $3.3\times$ difference that is further compounded by a

higher background ESRD rate in age- and health-matched Black non-donors.¹⁴

Age at donation

The life-expectancy cost decreased monotonically with older age at donation, from 56.1 days at age 25 to 19.6 days at age 55 (Figure 2). This gradient reflects the shorter remaining lifetime over which excess ESRD risk accumulates: younger donors have more life-years at risk and thus more opportunity for the elevated ESRD hazard to materialize.

Sex

Male donors faced a larger life-expectancy cost (49.4 days) than female donors (33.5 days), consistent with the male-sex within-donor ESRD hazard ratio of 1.88 reported by Massie et al.⁵ Sex and race effects compound: a Black male donor at age 40 loses 105 days, roughly double the 50 days for an overall male and four and a half times the 23 days for a White female donor of the same age (Figure 2). At age 25, Black male donors lose 272 days versus 32 days for White female donors.

Probabilistic sensitivity analysis

The PSA of 200 parameter draws (Figure 3) produced a mean Δ LE of +25.8 days (95% credible interval −1,091.8 to +1,064.6 days; P[donation beneficial] = 48%). The wide credible interval and near-50% probability of benefit are driven almost entirely by uncertainty in the donor all-cause mortality hazard ratio, which produces a tight cluster near −41 days when the HR is near its point estimate of 1.0 and a heavy left tail when the HR is drawn from the upper tail of the log-normal prior. Conditional on the HR remaining at 1.0, the value supported by the US-based literature and the O’Keeffe et al. meta-analysis,⁶ the model result is robustly and consistently negative.

One-way sensitivity analysis

The donor all-cause mortality hazard ratio dominated all other parameters by an order of magnitude (Table 1, Figure 4). Setting HR = 1.30⁷ yielded Δ LE = −1,069.9 days, a swing of more than 1,000 days from base case. No other parameter produced a swing exceeding 35 days. Dialysis mortality ($\pm 50\%$) shifted Δ LE by up to 15 days.

Donor priority benefit

Although KAS250 affords prior living donors (PLD) priority access to the deceased-donor waitlist,¹⁰ the resultant life-expectancy benefit to the donor is small at the population level. Removing PLD priority access entirely, setting the donor waitlist to the standard 1,765-day mean wait, worsened Δ LE by 0.7 days relative to base case. Reducing the priority wait to 50 days (median, equivalent to a mean of ≈ 72 days) had negligible additional effect. This insensitivity to the exact priority wait time arises because fewer than 0.3% of donors develop ESRD within 15 years and fewer still ever reach the transplant waitlist; the priority pathway is simply too rarely invoked to move a population-level outcome meaningfully.

Voucher benefit

The life-expectancy benefit of a kidney-donation voucher to a healthy, non-donor family member peaked at +0.6 days for a recipient designated at age 25, falling to +0.2 days at age 40 and becoming negligible by age 60. These small figures reflect the same dilution mechanism: conditional on developing ESRD and reaching the priority waitlist, priority access adds a clinically substantial +539 days of life expectancy for a patient whose ESRD onset occurs at age 40, and +109 days for onset at age 60 (Figure 5). However, since only approximately 0.04% of non-donors develop ESRD within 15 years, this large conditional benefit compresses to well under one day at the population level. Voucher benefit was modestly higher for Black non-donor family members (+0.9 days at age 40), owing to their approximately four-fold higher baseline ESRD incidence.¹⁴

Discussion

The mean life-expectancy cost to the individual donor (≈ 41 days) must be weighed against the benefit conferred on the recipient and, through the voucher mechanism, on designated family members who may one day develop ESRD. The ESRD-conditional model demonstrates that priority waitlist access alone is worth more than 200 days to a patient who develops ESRD at age 55, and approximately 110 days for onset at age 60; the full benefit of receiving a functioning kidney rather than remaining on long-term haemodialysis is larger still. Viewed in this frame, donation represents a transfer of life expectancy from donor to recipient on highly favourable terms: the donor foregoes approximately six weeks of expected life in exchange for providing the recipient with the prospect of years of additional survival.

This aggregate figure, however, conceals substantial heterogeneity that bears directly on informed consent. The life-expectancy cost of donation is three times larger for Black donors than for White donors, and nearly three times larger at age 25 than at age 55; a young Black donor and an older White donor are, in effect, being asked to accept quantitatively different bargains under the same generic counseling framework. Because this disparity traces to a measurable, race-specific difference in post-donation ESRD incidence rather than to any modifiable feature of the transplant episode itself, donor evaluation and consent discussions arguably warrant demographic-specific life-expectancy estimates rather than a single population-average number.

The model also reframes what priority-access policy is actually protecting. KAS250 and the NKR voucher mechanism are designed around a rare event, an eventual ESRD diagnosis in a prior donor or designated family member, and this analysis shows that their unconditional, population-level life-expectancy value is correspondingly small (under one day for a voucher holder, under one day for donor PLD priority) simply because so few beneficiaries ever draw on it. This is not evidence that such policies are unimportant: conditional on the rare event they are designed for, priority access is worth 110 to over 500 days at the individual level (depending on age at ESRD onset), and the dilution reflects low incidence rather than low

value per event. The appropriate way to communicate these mechanisms may therefore be closer to catastrophic insurance, whose expected value is small precisely because it is rarely invoked, than to an intervention expected to move average outcomes.

Several limitations temper these conclusions:

- The results are dominated by a single structural uncertainty, the donor all-cause mortality hazard ratio, for which the largest available meta-analysis reports no elevation in mortality risk while the longest-follow-up single cohort reports a hazard ratio of 1.30; the qualitative sign of the donor's life-expectancy cost is robust to this uncertainty, but its magnitude is not, and the true value likely depends on donor-recipient relatedness and selection practices that differ across health systems and were not modelled here.
- The model reports only undiscounted life-years and does not incorporate quality-of-life weighting; because dialysis is associated with substantially reduced health utility, a QALY-adjusted analysis would likely widen, not narrow, the donor's disadvantage during any dialysis-dependent interval.
- Several parameters were necessarily borrowed across source populations under an explicit sourcing rule (e.g., within-donor hazard ratios from Massie et al. applied to the Muzaale matched-control baseline), and no single cohort traces the complete pathway from living donation through ESRD onset to transplantation, so the calibration checks reported here constitute face validity rather than external validation against a purpose-built dataset.

Future work should prioritize prospective donor cohorts with sufficient follow-up to estimate all-cause mortality directly.

Acknowledgments and Statements and Declarations

Ethical Considerations

This study analyzed only publicly available, de-identified, aggregate data from published cohort studies and national registry reports (USRDS, SRTR/OPTN Annual Data Reports); no primary human-subjects data were collected. Institutional Review Board approval was therefore not applicable.

Consent to Participate

Not applicable. This study did not involve primary data collection from human participants.

Consent for Publication

Not applicable.

Declaration of Conflicting Interests

The author is a living kidney donor; that experience motivated this study. The author declares no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data Availability

All parameter sources are published, publicly available studies and registry reports, cited throughout the manuscript. The simulation code, parameter-assembly pipeline, and calibration checks are implemented in Python 3.10 (NumPy, pandas, matplotlib) and openly available in a public repository at <https://github.com/rupumped/kidney-donor-le>, consistent with Level 1 of the Transparency and Openness Promotion (TOP) Guidelines. No restricted-access data were used.

Demographic Classification Variables

Race is used in this analysis as reported in the underlying data sources rather than as a directly measured biological quantity. Donor-side race data^{1,4,5,15,16} derive from categories reported to the Organ Procurement and Transplantation Network (OPTN) at donor evaluation and registration; the non-donor comparator population used by⁴ was drawn from NHANES III, in which race is self-reported by participants. Race is included in the model because it is the variable available in these source populations and because it is strongly associated with measured post-donation ESRD incidence. Part of this association has an identified biological correlate, *APOL1* renal-risk-variant genotypes, which are more common among individuals of recent African ancestry and are directly implicated in donor kidney-disease risk.⁸ However, socioeconomic status, healthcare access, comorbidity burden, and other factors correlated with self-reported race in the United States were not available in the source registries and are not separately modelled here. The race-stratified life-expectancy estimates reported in this paper should therefore be interpreted as reflecting the net effect of these unmeasured, race-correlated factors together with any effect of race itself, rather than race operating as an isolated causal variable.

References

1. Lentine KL, Smith JM, Lyden GR et al. OPTN/SRTR 2023 annual data report: Kidney. *American Journal of Transplantation* 2025; 25(2, Supplement 1): S22–S137. DOI:<https://doi.org/10.1016/j.ajt.2025.01.020>. URL <https://www.sciencedirect.com/science/article/pii/S1600613525000267>. OPTN/SRTR 2023 Annual Data Report.
2. Cooper M, Kandaswamy R, Shannon P et al. A 100-year simulation of the National Kidney Registry's voucher program. *American Journal of Transplantation* 2026; 26(3): 587–595. DOI:<https://doi.org/10.1016/j.ajt.2025.11.006>. URL <https://www.sciencedirect.com/science/article/pii/S1600613525031053>.
3. Segev DL, Muzaale AD, Caffo BS et al. Perioperative mortality and long-term survival following live kidney donation. *JAMA* 2010; 303(10): 959–966. DOI:10.1001/jama.2010.237. URL <https://doi.org/10.1001/jama.2010.237>.

- https://jamanetwork.com/journals/jama/articlepdf/185508/joc05016_959_966.pdf.
4. Muzaale AD, Massie AB, Wang MC et al. Risk of end-stage renal disease following live kidney donation. *JAMA* 2014; 311(6): 579–586.
 5. Massie AB, Muzaale AD, Luo X et al. Quantifying postdonation risk of ESRD in living kidney donors. *Journal of the American Society of Nephrology: JASN* 2017; 28(9): 2749.
 6. O’keeffe LM, Ramond A, Oliver-Williams C et al. Mid- and long-term health risks in living kidney donors. *Annals of Internal Medicine* 2018; 168(4): 276–284. DOI:10.7326/M17-1235. URL <https://doi.org/10.7326/M17-1235>. PMID: 29379948, <https://doi.org/10.7326/M17-1235>.
 7. Mj  en G, Hallan S, Hartmann A et al. Long-term risks for kidney donors. *Kidney International* 2014; 86(1): 162–167. DOI:<https://doi.org/10.1038/ki.2013.460>. URL <https://www.sciencedirect.com/science/article/pii/S0085253815302489>.
 8. Genovese G, Friedman DJ, Ross MD et al. Association of trypanolytic APOL1 variants with kidney disease in African Americans. *Science* 2010; 329(5993): 841–845. DOI: 10.1126/science.1193032. URL <https://www.science.org/doi/abs/10.1126/science.1193032>. <https://www.science.org/doi/pdf/10.1126/science.1193032>.
 9. Matas AJ and Hays RE. Gender disparities and financial barriers to living kidney donation. *Journal of the American Society of Nephrology: JASN* 2018; 29(4): 1081.
 10. Wainright J, Kucheryavaya A, Klassen D et al. The impact of the new kidney allocation system on prior living kidney donors’ access to deceased donor kidney transplants: An early look. *American Journal of Transplantation* 2017; 17(4): 1103–1111. DOI:<https://doi.org/10.1111/ajt.14102>. URL <https://www.sciencedirect.com/science/article/pii/S1600613522249491>.
 11. Potluri V, Harhay MN, Wilson FP et al. Kidney transplant outcomes for prior living organ donors. *Journal of the American Society of Nephrology: JASN* 2014; 26(5): 1188.
 12. Cooper M, Leiser DB, Flechner SM et al. Ensuring the need is met: A 50-year simulation study of the National Kidney Registry’s family voucher program. *American Journal of Transplantation* 2021; 21(3): 1128–1137. DOI:<https://doi.org/10.1111/ajt.16101>. URL <https://www.sciencedirect.com/science/article/pii/S1600613522084477>.
 13. Murray CJ, Ezzati M, Flaxman AD et al. GBD 2010: design, definitions, and metrics. *The Lancet* 2012; 380(9859): 2063–2066.
 14. Grams ME, Sang Y, Levey AS et al. Kidney-failure risk projection for the living kidney-donor candidate. *New England Journal of Medicine* 2016; 374(5): 411–421. DOI:10.1056/NEJMoal510491. URL <https://www.nejm.org/doi/full/10.1056/NEJMoal510491>. <https://www.nejm.org/doi/pdf/10.1056/NEJMoal510491>.
 15. Johansen KL, Gilbertson DT, Li S et al. US Renal Data System 2024 annual data report: epidemiology of kidney disease in the United States. *American Journal of Kidney Diseases* 2025; 85(6): A8–A11.
 16. Johansen KL, Gilbertson DT, Li S et al. US Renal Data System 2025 annual data report: epidemiology of kidney disease in the United States. *American Journal of Kidney Diseases* 2026; 87(5): A6–A9.
 17. Sonnenberg FA and Beck JR. Markov models in medical decision making: A practical guide. *Medical Decision Making* 1993; 13(4): 322–338. DOI:10.1177/0272989X9301300409. URL <https://doi.org/10.1177/0272989X9301300409>. PMID: 8246705, <https://doi.org/10.1177/0272989X9301300409>.
 18. Punjala SR, Logan AJ, Brock GM et al. Determinants of long waiting time to kidney transplantation. *Transplantation Proceedings* 2024; 56(8): 1740–1751. DOI:<https://doi.org/10.1016/j.transproceed.2024.08.010>. URL <https://www.sciencedirect.com/science/article/pii/S0041134524004470>.
 19. Muzaale AD, Massie AB, Kucirka LM et al. Outcomes of live kidney donors who develop end stage renal disease. *Transplantation* 2016; 100(6): 1306.
 20. Arias E, Xu J and Kochanek K. United States life tables, 2021. *National Vital Statistics Reports* 2023; 72(12). URL <https://stacks.cdc.gov/view/cdc/132418>.
 21. Mid- and long-term health risks in living kidney donors. *Annals of Internal Medicine* 2018; 169(4): 265–266. DOI: 10.7326/L18-0341. URL <https://doi.org/10.7326/L18-0341>. PMID: 30128524, <https://doi.org/10.7326/L18-0341>.

Tables

Figures

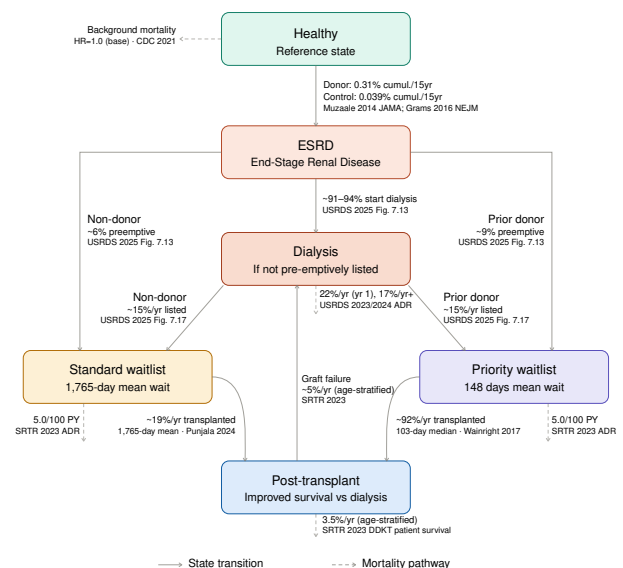


Figure 1. The Markov model for kidney donation.

Table 1. One-way sensitivity analysis: parameter values tested and the resulting change in life expectancy (donor arm minus non-donor arm), age 40 at donation, all other parameters held at base-case values. Negative Δ LE values indicate a net life-expectancy cost of donation relative to the matched non-donor arm under the value tested; positive values indicate a donor-arm benefit. The “8× controls (base case)” row reproduces the overall base-case result reported in the main results (Δ LE = −40.4 d).

Parameter (value tested)	Δ LE (days)	vs. base (days)
<i>ESRD relative risk, donor vs. non-donor (base case: 8×, 30.8/10,000)</i>		
2× controls	−5.7	+34.7
4× controls	−17.5	+22.9
8× controls (base case)	−40.4	0.0
11× controls (Mjøen upper bound)	−58.6	−18.2
<i>Standard waitlist mean wait (base case: 1,765 d)</i>		
24 months (730 d)	−40.5	−0.1
61 months (1,857 d, pre-KAS250)	−40.4	0.0
<i>Priority (PLD) waitlist mean wait (base case: $\approx$ 148 d, from 102.6 d median via $\mu = \text{median}/\ln 2$)</i>		
50 d median ($\approx$ 72 d mean, optimistic)	−40.4	0.0
200 d median ($\approx$ 289 d mean)	−40.5	−0.1
No priority access (PLD = standard, 1,765 d mean)	−41.1	−0.7
<i>Dialysis annual mortality (base case: 17%)</i>		
−50% (8.5%)	−25.7	+14.7
+50% (25.5%)	−47.8	−7.4
<i>Donor all-cause mortality HR (base case: 1.0)</i>		
HR = 1.30 (Mjøen pessimistic)	−1,069.9	−1,029.5
<i>Post-transplant survival (base case: DDKT quality)</i>		
LDKT quality (SRTR Figure KI 76)	−32.4	+8.0

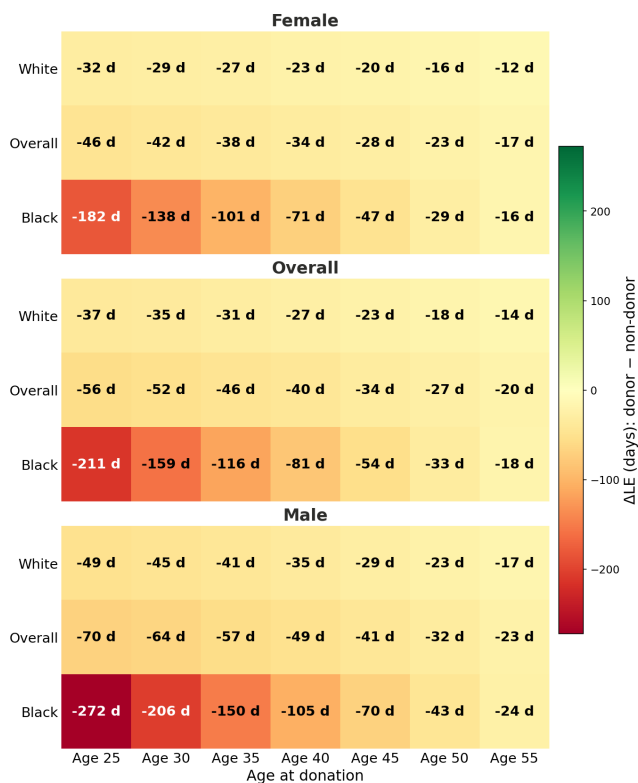


Figure 2. Life-expectancy impact of kidney donation by age, race, and sex. Each cell shows Δ LE (donor minus non-donor, days) for the indicated sex panel, race row, and age-at-donation column. Subgroup effects sum: a Black male donor at age 25 loses 272 days, compared with 12 days for a White female donor at age 55.

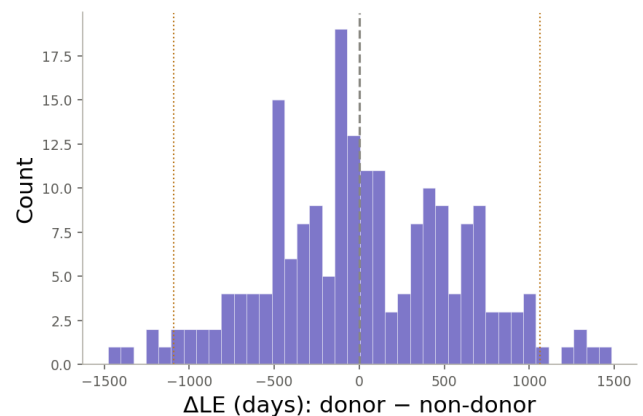


Figure 3. Probabilistic sensitivity analysis (PSA). Histogram of Δ LE (donor minus non-donor, days). The dashed vertical line marks zero; orange dotted lines mark the 95% credible interval bounds.

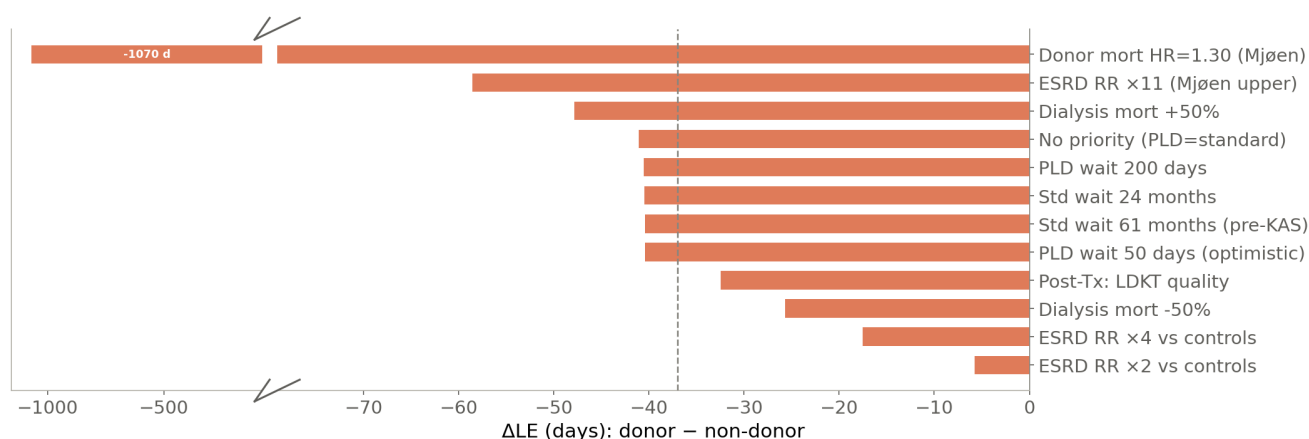


Figure 4. One-way sensitivity analysis — tornado chart. Each bar shows the Δ LE resulting from testing the indicated parameter value (see Table 1 for exact values). Vertical dashed line shows base case. The donor all-cause mortality HR (top bar, truncated at left) produces a $\approx 1,000$ -day shift that dwarfs every other parameter. The right panel uses a finer scale to show that all other parameters produce swings of ≤ 35 days. Priority waitlist parameters (“No priority” and “PLD wait” rows) are barely distinguishable from the base case, confirming the negligible population-level effect of transplant priority.

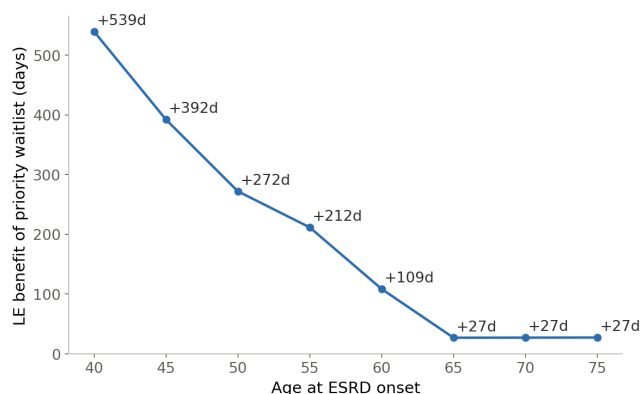


Figure 5. Priority waitlist LE benefit conditional on ESRD onset, by age. Δ LE (priority minus standard waitlist, days) for individuals who have already developed ESRD, as a function of age at ESRD onset. Conditional on ESRD, priority access is highly valuable: +539 days at age 40 onset, +109 days at age 60. This large conditional benefit is diluted to under 1 day at the population level by the low ESRD incidence in non-donors ($\approx 0.04\%/15$ yr) and in donors ($\approx 0.3\%/15$ yr).